Lab 8: Plate Tectonics

In this lab, we will be using Python and Google Earth Pro to explore aspects of plate tectonics. **Tectonics** is the branch of earth science that deals with the broad structural and deformational features of the outer part of the Earth, their origins, and the relationships between them.

Purpose: To gain familiarity with *the distribution of earthquakes and volcanoes around the Earth* in the context of the global distribution of topography and plate boundaries.

Answer all questions below. Include maps and other diagrams as it is useful or necessary for your answer, whether or not it is specifically requested. **Be sure to label all figures with a title and caption.**

Connections to Earthquakes and Volcanism (40 points)

Think about the geographic distribution of earthquakes and the relationship between earthquakes, topography, and plate boundaries. You will observe that earthquakes are not distributed randomly and generally they are not geographically isolated. **For most earthquakes, there is a systematic geographic pattern**.

Let's investigate the global distribution of earthquakes using Python. For this assignment, all coding questions will be answered in your copy of the provided Jupyter Notebook whereas all written questions will be answered here.

To start, read through the first two sections of the provided Jupyter Notebook. These sections introduce querying public APIs for data in Python and plotting data on maps respectively.

1. Look at your results from Coding Question 1. **What was the largest earthquake that occurred during the year 2011. What other major natural disaster accompanied it?** (You can look up the latter, just cite your source!)

Now, let's take a broader look at the global distribution of earthquakes.

2. Query the USGS database for all earthquakes with magnitudes greater than 5.5 between 2005 and 2025. Once the dataset loads, plot them on a map of the Earth with their color based on the depth and their size based on the magnitude. **Include a screenshot of the map in your report and comment on the distribution of earthquakes globally**.

3. Describe the general features of the earthquakes that you observe at each type of plate boundary (convergent/divergent/transform). **What types of earthquakes (i.e. shallow/deep/intermediate) are associated with each and what are their magnitudes and spatial distributions?**

Consider the figure on the following page showing the distribution of global volcanoes. Compare this to your map of global earthquakes.

4. Describe where you observe volcanoes, particularly around the rim of the Pacific Ocean. **How well do they match up with the earthquake distribution? With which type of plate boundary are volcanoes most often associated?**

5. Consider the clusters of volcanism in East Africa and Indonesia. **What are the general geologic processes causing each of these groups of volcanoes?** (*Hint*: there are differences in these two regions!). You are welcome to Google additional information, but remember that if you use sources other than the lecture or textbook be sure to cite them.

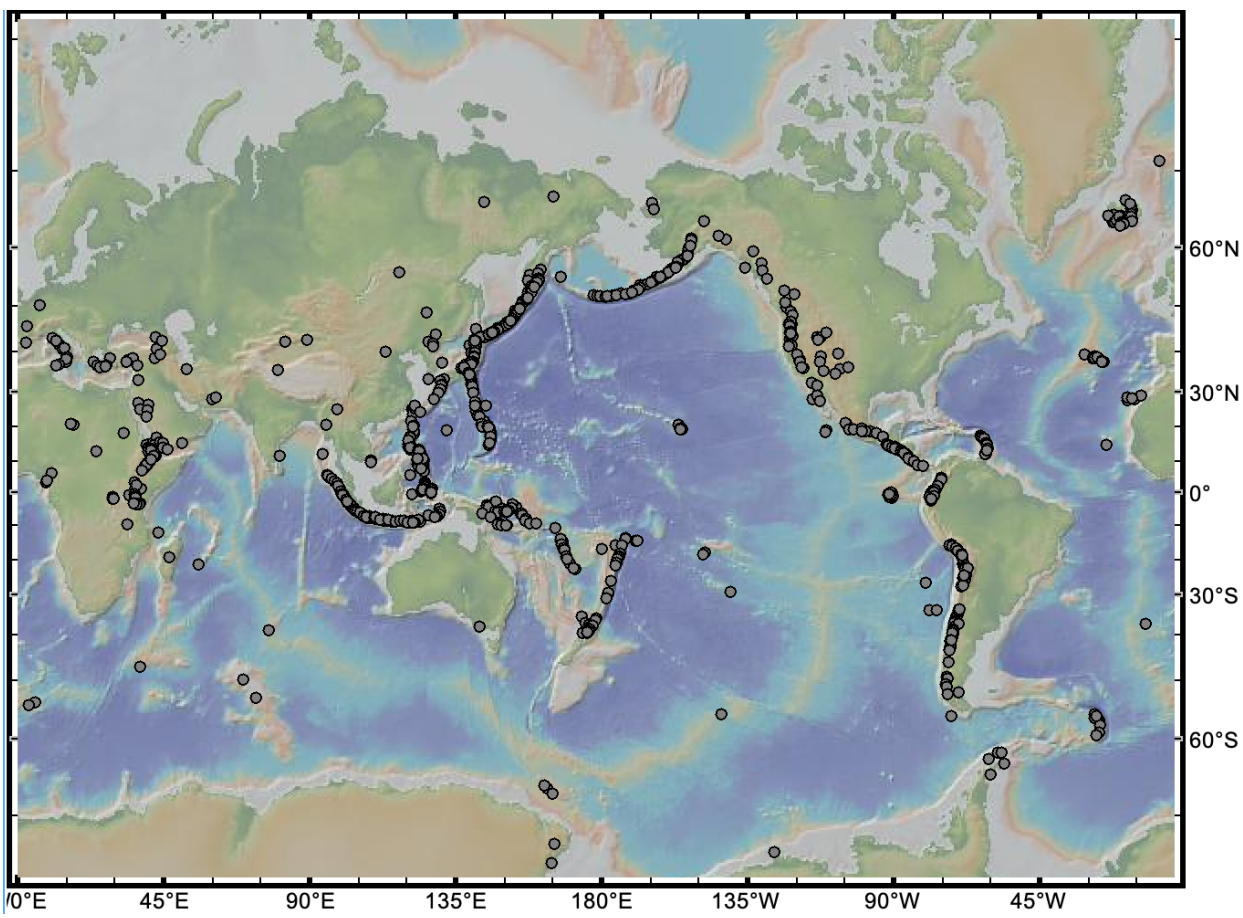

Figure 1: Locations of Global Volcanoes

Returning to your earthquake map in Python, adjust the latitude and longitude bounds for your map to focus on Eastern Europe and Asia. Note that earthquakes are distributed more broadly across the continent than, say, across the Pacific plate or the North American plate.

6. **What is the main geographic feature associated with the large area of broadly distributed earthquakes?**

To help you answer the above, we're now going to turn to Google Earth Pro and look at some elevation profiles. Navigate to this same region and draw two North-South paths across two separate areas where you observe these patches of distributed seismicity. Then, take a look at their elevation profiles.

7. **Include your two elevation profiles along with an image of the map showing the path of each. Label the areas along your path of active earthquakes and those with little to no associated seismicity.** (*This can be a visual estimate when you compare your two maps, but a trend/pattern should appear!*).

Let's return one last time to our earthquake map in Python. In our map, we color-coded our earthquakes by depth. Take a look at your map and find an area of deep earthquakes (>500 km deep) and adjust your latitude and longitude ranges to zoom in on this region. Note that these are nearly always close to intermediate depth and shallow earthquakes.

8. **What type of plate boundaries are these associated with? How deep do they get (approximately)? What is happening that causes the deep earthquakes and why are they only found in this context?** Include a figure or two as necessary to support your answer.